

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	AIML-A11
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: Intelligent Cyber Threat Detection System using Machine Learning

Abstract

Background and Problem Statement:

The rapid growth of digital technologies, cloud computing, Internet of Things (IoT), and online services has significantly increased the volume of network traffic generated every day. At the same time, cyber threats such as Distributed Denial-of-Service (DDoS) attacks, brute-force attacks, malware communication, ransomware, phishing, and network intrusions have become more sophisticated and frequent. Traditional cybersecurity solutions, including firewalls and signature-based Intrusion Detection Systems (IDS), primarily rely on predefined rules and known attack signatures. As a result, they often fail to detect new, unknown, or evolving cyber threats. Furthermore, manually monitoring millions of network packets generated every second is practically impossible, making automated and intelligent threat detection an essential requirement for modern organizations.

Proposed Solution:

This project proposes an *Intelligent Cyber Threat Detection System using Machine Learning* that provides real-time monitoring and analysis of network traffic to identify malicious activities. The proposed system captures and processes network traffic by extracting important features such as protocol type, packet size, connection duration, source and destination behavior, and traffic flow characteristics. These features are then analyzed using Machine Learning algorithms such as Random Forest, Support Vector Machine (SVM), XGBoost, or Neural Networks to classify network traffic as either normal or malicious. The system is capable of detecting multiple categories of cyber attacks, including DDoS attacks, brute-force attacks, malware communication, and port scanning with high accuracy.

Once suspicious activity is detected, the system immediately generates alerts and displays the detected threats through an interactive dashboard, enabling security administrators to respond quickly and effectively. All detected events are securely stored in a database for future investigation, reporting, and forensic analysis. The system also provides statistical visualizations of attack patterns, traffic distribution, and historical records, allowing organizations to gain valuable insights into their network security.

The primary objective of this project is to develop an intelligent, scalable, and automated cyber defense solution that enhances detection accuracy while minimizing false alarms. The system can be trained and evaluated using publicly available cybersecurity datasets such as CICIDS2017, UNSW-NB15, and NSL-KDD to ensure reliable performance across various attack scenarios. By integrating Artificial Intelligence with cybersecurity, the proposed solution offers a proactive approach to identifying threats before they cause significant damage. This project has potential applications in enterprise networks, cloud computing environments, educational institutions, government organizations, financial sectors, and IoT infrastructures, contributing to stronger cybersecurity and improved protection of digital assets.

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References (Category specific)

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Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization.
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2. Tavallaee, M., Bagheri, E., Lu, W., & Ghorbani, A. A. (2009).
A Detailed Analysis of the KDD CUP 99 Dataset.
IEEE Symposium on Computational Intelligence for Security and Defense Applications.
3. Moustafa, N., & Slay, J. (2015).
UNSW-NB15: A Comprehensive Data Set for Network Intrusion Detection Systems.
Military Communications and Information Systems Conference (MilCIS).
4. Breiman, L. (2001).
Random Forests.
Machine Learning Journal, 45(1), 5–32.
5. Chen, T., & Guestrin, C. (2016).
XGBoost: A Scalable Tree Boosting System.
Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining.
6. Scikit-learn Developers.
Scikit-learn: Machine Learning in Python.
Available: <https://scikit-learn.org/>
7. TensorFlow Documentation.
Machine Learning Framework for Deep Learning Applications.
Available: <https://www.tensorflow.org/>
8. Wireshark Foundation.
Wireshark Network Protocol Analyzer Documentation.
Available: <https://www.wireshark.org/>
9. CICIDS2017 Dataset.
Canadian Institute for Cybersecurity Intrusion Detection Evaluation Dataset.
Available: <https://www.unb.ca/cic/datasets/ids-2017.html>
10. National Institute of Standards and Technology (NIST).
NIST Cybersecurity Framework (CSF) 2.0.
Available: <https://www.nist.gov/cyberframework>

Project Team Members

S. No.	Regd. No.	Name of the Student	Signature
01	23BQ1A6142	G Vyshnavi	
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Project Guide

Name: Mrs K Sireesha

Signature:

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INSTRUCTIONS

(No need to submit this page so you don't have to take the print)

Project category

Project categories were mentioned only for the sake of documentation. **All of you must chose 'Research oriented' projects only.** Very few projects would be accepted if the project statements are chosen from Smart India Hackathon (SIH) or from any other reputed platform for that matter. In such cases, you have to write the problem ID mentioned in those platforms as it is in the '**Project category**' section. Also provide the link to the problem statement in the '**References**' section.

To write Abstract

(Use the following as a mental checklist)

- Background & Problem Statement:** What is the exact real-world problem your project addresses? State it precisely and narrowly not "we are improving healthcare" but "patients in rural areas lack timely access to dermatology consultation."
- Target Users / Stakeholders:** Who exactly faces this problem? A defined persona, not "everyone" e.g. "first-generation college students applying for scholarships."
- Existing Gap & Evidence:** What do people currently do to cope with this problem, and why is it inadequate? Cite any survey, interview, article, or report that confirms the problem is real.
- Related Work / Competitive Landscape:** What are the existing tools/apps/products/papers that partially address this, and how will yours differ or improve on them? (Feeds directly into your References in Section 3.)
- Proposed Approach (overview):** One paragraph describing your approach/technique/system at a high level no implementation detail needed yet.
- Scope & Boundaries:** What will you solve, and explicitly what will you NOT attempt within this project's timeline?
- Expected Outcome / Success Criteria:** How will you know the project succeeded? State one or two measurable, testable outcomes (e.g. "90% of test users complete the task without guidance").

To write category specific references

Research oriented	Cite 1 base paper your problem statement builds on. The paper must be published within the last 3 years (2023 onward as of 2026). Add 2-4 supporting papers from the literature review if relevant.
Application oriented	Identify 2-3 existing, similar applications to use as your benchmark. For each, cite the app/website itself (as a web reference) and, if available, any article or documentation describing it.
Product oriented	Identify 2-3 existing, similar products (hardware/software) to use as your benchmark. Cite the manufacturer's page, datasheet, or a credible review/comparison article for each.